

1.2.7:

Solution: We must show that $f = g$ in $\mathcal{F}(S, \mathbb{R})$ for $S = \{0, 1\}$ and $f(t) = 2t + 1$ and $g(t) = 1 + 4t - 2t^2$. It will suffice to show $f(0) = g(0)$ and $f(1) = g(1)$:

$$f(0) = 2(0) + 1 = 1 = 1 + 4(0) - 2(0)^2 = g(0)$$

$$f(1) = 2(1) + 1 = 3 = 1 + 4(1) - 2(1)^2 = g(1)$$

Thus $f = g$.

We must now show $f + g = h$ for $h(t) = 5^t + 1$. Once again we have:

$$(f + g)(0) = f(0) + g(0) = 1 + 1 = 2 = 5^0 + 1 = h(0)$$

$$(f + g)(1) = f(1) + g(1) = 3 + 3 = 6 = 5^1 + 1 = h(1)$$

1.2.8:

Solution: We must show that $(a + b)(x + y) = ax + by + bx + by$ for any $x, y \in V$ and $a, b \in \mathbb{F}$ for V a vector space. We have:

$$(a + b)(x + y) = (a + b)x + (a + b)y \quad (\text{VS7})$$

$$= ax + by + ay + by \quad (\text{VS8})$$

$$= ax + by + bx + by \quad (\text{VS1})$$

which was to be shown. Note that parenthesis are not needed on the RHS due to (VS2) which by repeated application allows us to defined the addition of any finite numbers of vectors unambiguously as any grouping of parenthesis is equivalent under vector addition in V . We state

$$(\text{VS2}): \forall \vec{x}, \vec{y}, \vec{z} \in V, (\vec{x} + \vec{y}) + \vec{z} = \vec{x} + (\vec{y} + \vec{z}).$$

Now, specifically for $w, x, y, z \in V$ we have

$$(w + x) + y = w + (x + y) \Rightarrow ((w + x) + y) + z = (w + (x + y)) + z$$

by (VS2) and then adding z to both sides. It then follows that

$$(w + (x + y)) + z = w + ((x + y) + z)$$

through (VS2) with $\vec{y} = (x + y)$. Then,

$$(x + y) + z = x + (y + z) \Rightarrow w + ((x + y) + z) = w + (x + (y + z))$$

once again by (VS2) and then adding w to both sides (from the left). Applying (VS2) with $\vec{x} = (w + x)$, we see

$$((w + x) + y) + z = (w + x) + (y + z)$$

so

$$\begin{aligned} ((w + x) + y) + z &= (w + (x + y) + z) \\ &= w + ((x + y) + z) \\ &= w + (x + (y + z)) \\ &= (w + x) + (y + z) \end{aligned}$$

and thus parenthesis are not needed. □

1.2.11:

Solution:

Proof (Corollary 1 of Theorem 1.1). Suppose that both $0, 0' \in V$ satisfy (VS3). Then $x + 0 = x \forall x \in V$ and also $x + 0' = x \forall x \in V$. Take $x = 0'$ in the first equation to get $0' + 0 = 0'$. Applying (VS3) with $0'$ we see $0' + 0 = 0' + 0'$. It follows from Theorem 1.1' that $0' = 0$ which was to be shown. $\square$

Proof (Corollary 2 of Theorem 1.1). Suppose that both y and y' satisfy (VS4) for $y, y' \in V$. Then, $x + y = \vec{0}$ and $x + y' = \vec{0}$ for $x \in V$. By Corollary 1, $\vec{0}$ is unique so we have $x + y = x + y'$ and thus $y = y'$ by Theorem 1.1'. $\square$

Proof (Theorem 1.2(c)). We must show $a\vec{0} = \vec{0}$ for any $a \in \mathbb{F}$. But,

$$a\vec{0} + a\vec{0} = a(\vec{0} + \vec{0}) \quad (\text{VS7})$$

$$= a\vec{0} \quad (\text{VS3})$$

$$= a\vec{0} + \vec{0} \quad (\text{VS3})$$

$$\Rightarrow a\vec{0} = \vec{0} \quad (\text{Thm1.1'})$$

which was to be shown. $\square$

1.2.11:

Solution: We must show $V = \{\vec{0}\}$ is a vector space over $\mathbb{F}$. We define $\vec{0} + \vec{0} = \vec{0}$ and $c\vec{0} = \vec{0}$ for $c \in \mathbb{F}$.

In order to show V is a vector space we must show that (VS1) - (VS8) hold true.

(VS1): (VS1) is immediate by definition of vector addition in V

(VS2): We have that $(\vec{0} + \vec{0}) + \vec{0} = \vec{0} + \vec{0} = \vec{0} = \vec{0} + \vec{0} = \vec{0} + (\vec{0} + \vec{0})$ and so (VS2) holds.

(VS3): Using the fact that V is a singleton set, $x = \vec{0}$ and $\theta = \vec{0}$ where θ is the additive identity in (VS3). Thus $x + \theta = \vec{0} + \vec{0} = \vec{0}$ by definition of vector addition, thus (VS3) holds.

(VS4): Once again, leveraging the fact that V is a singleton it must be the case that $x = y = \vec{0}$ and so by definition, $\vec{0} + \vec{0} = \vec{0}$ and (VS4) holds.

(VS5): By definition of scalar multiplication in V (VS5) is trivial as $1 \in \mathbb{F}$ and $c\vec{0} = \vec{0} \forall c \in \mathbb{F}$.

(VS6): Also using the definition of scalar multiplication and the fact that for $a, b \in \mathbb{F}, ab \in \mathbb{F}$ (by field axioms), $(ab)\vec{0} = \vec{0} = a(b\vec{0}) = a\vec{0} = \vec{0}$, showing (VS6) is also true.

(VS7): Similarly we see that for $a \in \mathbb{F}, a(\vec{0} + \vec{0}) = a\vec{0} + b\vec{0} = \vec{0}$ (again, since V is a singleton we may take $x = y = \vec{0}$ for (VS7)) showing (VS7).

(VS8): Finally, field axioms again tell us that for $a, b \in \mathbb{F}, a + b \in \mathbb{F}$ so $(a + b)\vec{0} = \vec{0} = a\vec{0} + b\vec{0}$ by definition of scalar multiplication.

Since (VS1)-(VS8) have been found to be true, we may now conclude that V is a vector space over $\mathbb{F}$. $\square$

1.2.12:

Solution: We must show $V = \{f : \mathbb{R} \rightarrow \mathbb{R} \mid f(t) = f(-t) \forall t \in \mathbb{R}\}$ is a vector space. First we must show that V is closed under vector addition and scalar multiplication. Consider $f, g \in V$, by definition of vector addition we have that

$$(f + g)(s) = f(s) + g(s) = f(-s) + g(-s) = (f + g)(-s)$$

and so $f + g \in V$. As well, consider some $c \in \mathbb{R}$ and $f \in V$, then

$$cf(s) = cf(-s)$$

and so $cf \in V$ as well. We now show that VS1 – VS8 hold.

(VS1): (VS1) is immediate by the fact that $f, g \in V$ are real valued when evaluated at some $c \in \mathbb{R}$ and addition in $\mathbb{R}$ is commutative.

(VS2): (VS2) is also immediate by the fact that $f, g, h \in V$ are real valued when evaluated at some arbitrary $c \in \mathbb{R}$ and addition in $\mathbb{R}$ is associative.

(VS3): We define the function $I : \mathbb{R} \rightarrow \mathbb{R}$ by $s \mapsto 0$ so that $I(s) = I(-s) = 0$ for any $s \in \mathbb{R}$. It follows that $I \in V$. We claim that I is the additive identity for V . Consider any $f \in V$ and notice

$$(f + I)(s) = f(s) + I(s) = f(s) + 0 = f(s)$$

where $0 \in \mathbb{R}$ is the additive identity of the real numbers. Thus, $I \in V$ and $x + I = x$ for all $x \in V$.

(VS4): Since V is closed under scalar multiplication, if $f \in V$ then $-1 * f = -f \in V$ so $(f + (-f))(s) = f(s) + (-f(s)) = 0$.

(VS5): As before, (VS5) is immediate since $f \in V$ is real when evaluated at any $c \in \mathbb{R}$ so $1 * f(c) = f(c)$.

(VS6): Also using the definition of scalar multiplication and the fact that for $a, b \in \mathbb{R}, ab \in \mathbb{R}$ (by field axioms), $(ab)f = a(bf)$.

(VS7): Similarly we see that for $c \in \mathbb{R}, f(c) \in \mathbb{R}$ so for $a \in \mathbb{R}$ and $f, g \in V$, $a(f + g)(s) = af(s) + ag(s)$ by field axioms

(VS8): Finally, field axioms again tell us that for $a, b \in \mathbb{R}, a + b \in \mathbb{R}$ so for $f \in V$ and some $c \in \mathbb{R}$, $(a + b)f(s) = af(s) + bf(s)$.

Because any $f \in V$ is $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(t) \in \mathbb{R}$ so (VS1)-(VS8) automatically hold due to the field axioms of $\mathbb{R}$.

1.2.17:

Solution: V **cannot** be a vector space since for $(a_1, a_2) \in V$, $1(a_1, a_2) = (a_1, 0) \neq (a_1, a_2)$ for $a_2 \neq 0$ which violates (VS5).

1.3.5:

Solution: We must show $A + A^t$ is symmetric for any square matrix A . It is equivalent to show that $(A + A^t)^t = A + A^t$. But,

$$(A + A^t)^t = A^t + (A^t)^t = A^t + A = A + A^t$$

which was to be shown.

1.3.8

Solution:

- (a) W_1 is **not** a subspace of $\mathbb{R}^3$. Take $c = (1, 0, 0) \in \mathbb{R}^3$ and any $\vec{x} = (a_1, a_2, a_3) \in W_1$. By definition of W_1 , $\vec{x} = (3a_2, a_2, -a_2)$ but $c\vec{x} = (3a_2, 0, 0) \notin W_1$, thus Thm1.3(c) is not true and W_1 cannot be a subspace of $\mathbb{R}^3$.
- (b) W_2 is **not** a subspace of $\mathbb{R}^3$ since $\vec{0} \notin W_2$ since $a_1 = a_3 + 2$ implies that $a_1 = 0$ and $a_3 = 0$ cannot be true at the same time and thus W_1 cannot be a subspace of $\mathbb{R}^3$.

1.3.11

Solution: W is a subspace of $P(\mathbb{F})$. By definition $\vec{0} \in W$. Also, for $x, y \in W, x+y \in W$ since adding two polynomials of equal degree will always result in another polynomial of exactly degree n or $\vec{0}$. Finally, for $c \in \mathbb{F}$ and $\vec{x} \in W, c\vec{x} \in W$ since scalar multiplication cannot change the degree of polynomial. Thus, by Thm1.3 W is a subspace of $P(\mathbb{F})$ for $n \geq 1$.

⁰LaTeX code for this document can be found on github [here](#)